

* Review

疊

$$1. a \sin x + b \cos x = \sqrt{a^2 + b^2} \left(\sin x \times \frac{a}{\sqrt{a^2 + b^2}} + \cos x \times \frac{b}{\sqrt{a^2 + b^2}} \right)$$
$$= \sqrt{a^2 + b^2} \sin(x + \theta), \text{ where } \begin{cases} \sin \theta = \frac{b}{\sqrt{a^2 + b^2}} \\ \cos \theta = \frac{a}{\sqrt{a^2 + b^2}} \end{cases}$$

$$2. a \cos x - b \sin x = \sqrt{a^2 + b^2} \left(\cos x \cdot \frac{a}{\sqrt{a^2 + b^2}} - \sin x \cdot \frac{b}{\sqrt{a^2 + b^2}} \right)$$
$$= \sqrt{a^2 + b^2} \cos(x + \theta), \text{ where } \begin{cases} \sin \theta = \frac{b}{\sqrt{a^2 + b^2}} \\ \cos \theta = \frac{a}{\sqrt{a^2 + b^2}} \end{cases}$$

§ 0.5

(What the inputs be)

$f(x)$ 的 domain = x 所有可在 $f(x)$ 取值的 set.

e.g. $f(x) = \frac{1}{x}$

$f: (-\infty, 0) \cup (0, \infty) \rightarrow \mathbb{R}$ $\mathbb{N}, \mathbb{Z}$

Dom: $\{x \mid x \neq 0\}$ or $\{x \mid x \in (-\infty, 0) \cup (0, \infty)\}$

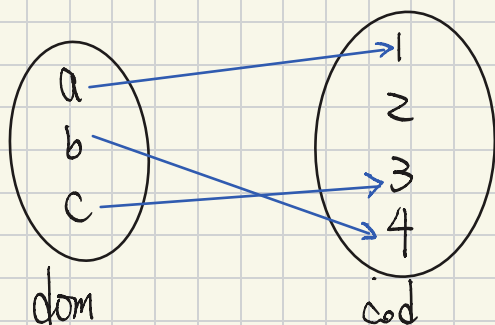
e.g. $f(x) = \sqrt{x}$

Dom: $\{x \mid x \geq 0\}$ or $\{x \mid x \in [0, \infty)\}$

Codomain: 所有可能的関数結果. (What the outputs can be)

Range: 所有實際上從 domain 對應到 codomain 的 set.
(What the output are.)

eg.



$$\text{domain} = \{a, b, c\}$$

$$\text{cod} = \{1, 2, 3, 4\}$$

$$\text{range} = \{1, 3, 4\}$$

eg. $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$

$$\text{domain} = \mathbb{R}$$

$$\text{cod} = \mathbb{R}$$

$$\text{range} = [0, \infty), \{y = f(x) \mid y \geq 0\}$$

eg. $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = \frac{1}{x}$

$$\text{dom} = \{x \mid x \neq 0\} \text{ or } (-\infty, 0) \cup (0, \infty)$$

$$\text{cod} = \mathbb{R}$$

$$\text{range} = \{y = f(x) \mid y \neq 0\} \text{ or } (-\infty, 0) \cup (0, \infty)$$



Functions 的平移和伸缩

1. 鉛直平移: if $c > 0$,

$\begin{cases} f(x) + c : \text{向上 } c \text{ units} \\ f(x) - c : \text{向下 } c \text{ units} \end{cases}$

2. 水平平移:

$\begin{cases} f(x-c) : \text{向右 } c \text{ units} \\ f(x+c) : \text{向左 } c \text{ units} \end{cases}$

3. 鉛直伸缩:

$cf(x) : f(x)$ 鉛直伸缩 c 倍

4. 水平伸缩:

$f(cx) : f(x)$ 水平伸缩 $\frac{1}{c}$ 倍

eg $f(2x-4)$ — 1. 先平移再伸缩: $f(x) \rightarrow f(x-4)$

$$\rightarrow f(2x-4)$$

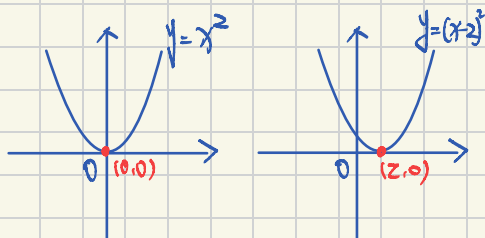
$\Rightarrow$ 右移 4 units, 再伸缩 $\frac{1}{2}$ 倍

2. 先伸缩再平移: $f(x) \rightarrow f(2x)$

$$\rightarrow f(2(x-2)) = f(2x-4)$$

$\Rightarrow$ 伸缩 $\frac{1}{2}$ 倍,

再右移 2 units.



$$x^2 \rightarrow (x-2)^2$$

$(0,0) \rightarrow (2,0)$

右移 2 units

37. Prove that, for some constant β ,

$$4 \cos x - 3 \sin x = 5 \cos(x + \beta).$$

Then, estimate the value of β .

§0.4

$$\begin{aligned} \text{pf } 4 \cos x - 3 \sin x &= \sqrt{4^2 + 3^2} \left(\cos x \cdot \frac{4}{\sqrt{4^2 + 3^2}} - \sin x \cdot \frac{3}{\sqrt{4^2 + 3^2}} \right) \\ &= 5 \left(\cos x \cdot \frac{4}{5} - \sin x \cdot \frac{3}{5} \right) \\ &= 5 \cos(x + \beta), \text{ where } \cos x = \frac{4}{5}, \sin x = \frac{3}{5}, \\ \beta &\approx 37^\circ (36.8698976^\circ). \end{aligned}$$

§0.5

In exercises 1–6, find the compositions $f \circ g$ and $g \circ f$, and identify their respective domains.

3. $f(x) = \frac{1}{x}, \quad g(x) = x^3 + 4$

4. $f(x) = \sqrt{1-x}, \quad g(x) = \tan x$

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ (g \circ f)(x) &= g(f(x)) \end{aligned}$$

3. $(f \circ g)(x) = f(g(x)) = \frac{1}{x^3 + 4} \quad x^3 + 4 \neq 0 \Rightarrow x^3 \neq -4 \Rightarrow x \neq (-4)^{\frac{1}{3}}$
 domain: $\{x \mid x \neq (-4)^{\frac{1}{3}}\} \quad (\text{or } (-\infty, (-4)^{\frac{1}{3}}) \cup ((-4)^{\frac{1}{3}}, \infty))$

$$(g \circ f)(x) = g(f(x)) = \left(\frac{1}{x}\right)^3 + 4$$

domain: $x^3 \neq 0 \Rightarrow x \neq 0 \Rightarrow \{x \mid x \neq 0\} \quad (\text{or } (-\infty, 0) \cup (0, \infty))$

4. $f(x) = \sqrt{1-x}, \quad g(x) = \tan x$
 $(f \circ g)(x) = f(g(x)) = \sqrt{1 - \tan x}$

dom: $\{x \mid \frac{\pi}{2} + k\pi < x \leq \frac{\pi}{4} + k\pi, k \in \mathbb{Z}\}$

$1-x \geq 0 \Rightarrow x \leq 1$

$(g \circ f)(x) = \tan \sqrt{1-x} \quad \sqrt{1-x} \neq \frac{\pi}{2} + k\pi, \quad 1-x \neq \left(\frac{\pi}{2} + k\pi\right)^2$
 dom: $\{x \mid x \leq 1 \text{ and } x \neq 1 - \left(\frac{\pi}{2} + k\pi\right)^2, k \in \mathbb{Z}\} \Rightarrow x \neq 1 - \left(\frac{\pi}{2} + k\pi\right)^2, k \in \mathbb{Z}$

11. 找 $f(x)$ and $g(x)$ such that $(f \circ g)(x) = (4x+1)^2 + 3$

$$f(x) = x^2 + 3$$

$$g(x) = 4x+1$$

12. $4(x+1)^2 + 3$

$$f(x) = 4x^2 + 3$$

$$g(x) = x+1$$

In exercises 37–42, complete the square and explain how to transform the graph of $y = x^2$ into the graph of the given function.

37. $f(x) = x^2 + 2x + 1$

38. $f(x) = x^2 - 4x + 4$

39. $f(x) = x^2 + 2x + 4$

40. $f(x) = x^2 - 4x + 2$

41. $f(x) = 2x^2 + 4x + 4$

42. $f(x) = 3x^2 - 6x + 2$

$$\begin{aligned} 42. \quad x^2 &\rightarrow \boxed{3x^2 - 6x + 2} = 3(x^2 - 2x) + 2 \\ &= 3(x-1)^2 + 2 - 3 \\ &= 3(x-1)^2 - 1 \end{aligned}$$

$3(x-1)^2 - 1$ 向右移 1 unit $\rightarrow$ 伸 $\frac{1}{3}$ 倍 $\rightarrow$ 下 1 unit

↑
水平
↑
向

Note: $\tan x$ 的 domain: $\{x \mid \frac{\pi}{2} + k\pi < x < \frac{3\pi}{2} + k\pi, k \in \mathbb{Z}\}$

也可以這樣寫 $\rightarrow$ or $(\frac{\pi}{2} + k\pi, \frac{3\pi}{2} + k\pi), k \in \mathbb{Z}$

* 1.2

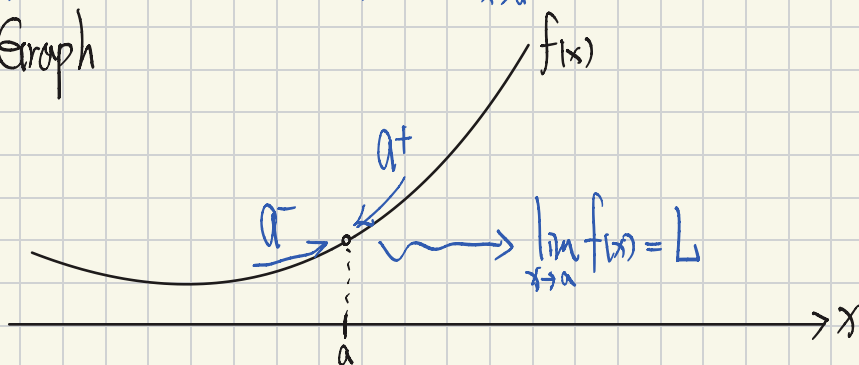
1.2: Graph to find limit and L-R side limit

① Left - Right side limit

Textbook P.54:

$$\lim_{x \rightarrow a} f(x) = L \in \mathbb{R} \Leftrightarrow \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$

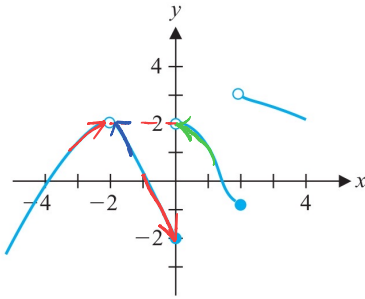
② Graph



$$2. \lim_{x \rightarrow -1} \frac{x^2 + x}{x^2 - x - 2} = \lim_{x \rightarrow -1} \frac{\cancel{x(x+1)}}{(x-2)\cancel{(x+1)}} = \lim_{x \rightarrow -1} \frac{x}{x-2} = \frac{-1}{-3} = \frac{1}{3}$$

$$5. \lim_{x \rightarrow 3} \frac{3x-9}{x^2-5x+6} = \lim_{x \rightarrow 3} \frac{3\cancel{(x-3)}}{(x-2)\cancel{(x-3)}} = \frac{3}{3-2} = 3$$

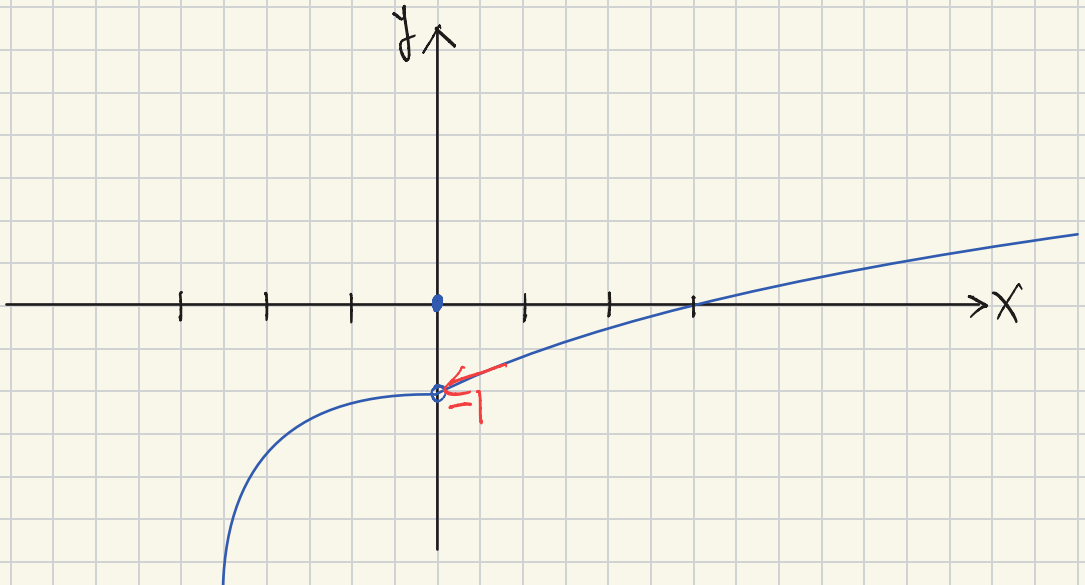
In exercises 7 and 8, identify each limit or state that it does not exist.



7. (a) $\lim_{x \rightarrow 0^-} f(x) = -2$ (b) $\lim_{x \rightarrow 0^+} f(x) = 2$ (c) $\lim_{x \rightarrow 0} f(x) = \text{DNE}$
 (d) $\lim_{x \rightarrow -2^-} f(x) = 2$ (e) $\lim_{x \rightarrow -2^+} f(x) = -2$ (f) $\lim_{x \rightarrow -2} f(x) = -2$
 (g) $\lim_{x \rightarrow -1} f(x)$ (h) $\lim_{x \rightarrow 1^-} f(x)$

10. Sketch the graph of $f(x) = \begin{cases} x^3 - 1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ \sqrt{x+1} - 2 & \text{if } x > 0 \end{cases}$ and identify each limit.

- (a) $\lim_{x \rightarrow 0^-} f(x) = -1$ (b) $\lim_{x \rightarrow 0^+} f(x) = -1$ (c) $\lim_{x \rightarrow 0} f(x) = -1$
 (d) $\lim_{x \rightarrow -1} f(x)$ (e) $\lim_{x \rightarrow 1^-} f(x)$



32. Give an example of a function f such that $\lim_{x \rightarrow 0} f(x)$ exists and $f(0)$ exists, but $\lim_{x \rightarrow 0} f(x) \neq f(0)$.

§1.4 Continuity

Exercise 10. $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = -1 \Rightarrow \lim_{x \rightarrow 0} f(x) = -1$, but $f(0) = 0$

A In exercises 13–22, use numerical and graphical evidence to conjecture whether $\lim_{x \rightarrow a} f(x)$ exists. If not, describe what is happening at $x = a$ graphically.

13. $\lim_{x \rightarrow 0} \frac{x^2 + x}{\sin x}$

14. $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x^2 - 2x + 1}$

15. $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$

16. $\lim_{x \rightarrow 0} x \csc 2x$

17. $\lim_{x \rightarrow 1} \frac{\sqrt{5-x} - 2}{\sqrt{10-x} - 3}$

18. $\lim_{x \rightarrow 0} \frac{x^2 + 4x}{\sqrt{x^3 + x^2}}$

19. $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$

20. $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$

21. $\lim_{x \rightarrow 2} \frac{x - 2}{|x - 2|}$

22. $\lim_{x \rightarrow -1} \frac{|x + 1|}{x^2 - 1}$

$$\begin{aligned} 17. \lim_{x \rightarrow 1} \frac{\sqrt{5-x} - 2}{\sqrt{10-x} - 3} &= \lim_{x \rightarrow 1} \frac{(5-x-4)(\sqrt{10-x}+3)}{(10-x-9)(\sqrt{5-x}+2)} \\ &= \lim_{x \rightarrow 1} \frac{\cancel{(5-x)}(\sqrt{10-x}+3)}{\cancel{(10-x)}(\sqrt{5-x}+2)} \\ &= \frac{\sqrt{10-1}+3}{\sqrt{5-1}+2} = \frac{3+3}{2+2} = \frac{6}{4} \end{aligned}$$

$$\begin{aligned} 18. \lim_{x \rightarrow 0} \frac{x^2 + 4x}{\sqrt{x^3 + x^2}} &= \lim_{x \rightarrow 0} \frac{\cancel{x}(x+4)}{\cancel{x}\sqrt{x+1}} \\ &= \frac{4}{1} = 4 \end{aligned}$$

$$21. \lim_{x \rightarrow 2} \frac{x-2}{|x-2|}$$

$$\lim_{x \rightarrow 2^-} \frac{x-2}{|x-2|} = \lim_{x \rightarrow 2^-} \frac{x-2}{-(x-2)} = -1$$

$$\lim_{x \rightarrow 2^+} \frac{x-2}{|x-2|} = \lim_{x \rightarrow 2^+} \frac{x-2}{x-2} = 1$$

$$\Rightarrow \lim_{x \rightarrow 2} \frac{x-2}{|x-2|} = \text{DNE}$$

$$22. \lim_{x \rightarrow -1} \frac{|x+1|}{x^2-1} = \lim_{x \rightarrow -1} \frac{|x+1|}{(x+1)(x-1)}$$

$$\lim_{x \rightarrow -1^-} \frac{|x+1|}{(x+1)(x-1)} = \lim_{x \rightarrow -1^-} \frac{-(x+1)}{(x+1)(x-1)} = \lim_{x \rightarrow -1^-} \frac{-1}{x-1} = \frac{-1}{-2} = \frac{1}{2}$$

$$\lim_{x \rightarrow -1^+} \frac{|x+1|}{(x+1)(x-1)} = \lim_{x \rightarrow -1^+} \frac{(x+1)}{(x+1)(x-1)} = \lim_{x \rightarrow -1^+} \frac{1}{x-1} = \frac{-1}{2}$$

$$\text{Since } \lim_{x \rightarrow -1^-} \frac{|x+1|}{x^2-1} \neq \lim_{x \rightarrow -1^+} \frac{|x+1|}{x^2-1}, \lim_{x \rightarrow -1} \frac{|x+1|}{x^2-1} \text{ DNE.}$$

考試注意事項：

1. 使用沒有在題目定義的符號，需要說明。
2. 定義域(domain)、對應域(codomain)和值域(range)請用集合表示。
區間(intervals)是集合可以直接寫。
3. 極限題不能使用羅必達法則(L'Hopital's Rule)和泰勒展開式(Taylor's Expansion)，使用直接0分。

補充：range and codomain 用 $\{y = f(x)\}$ 表示